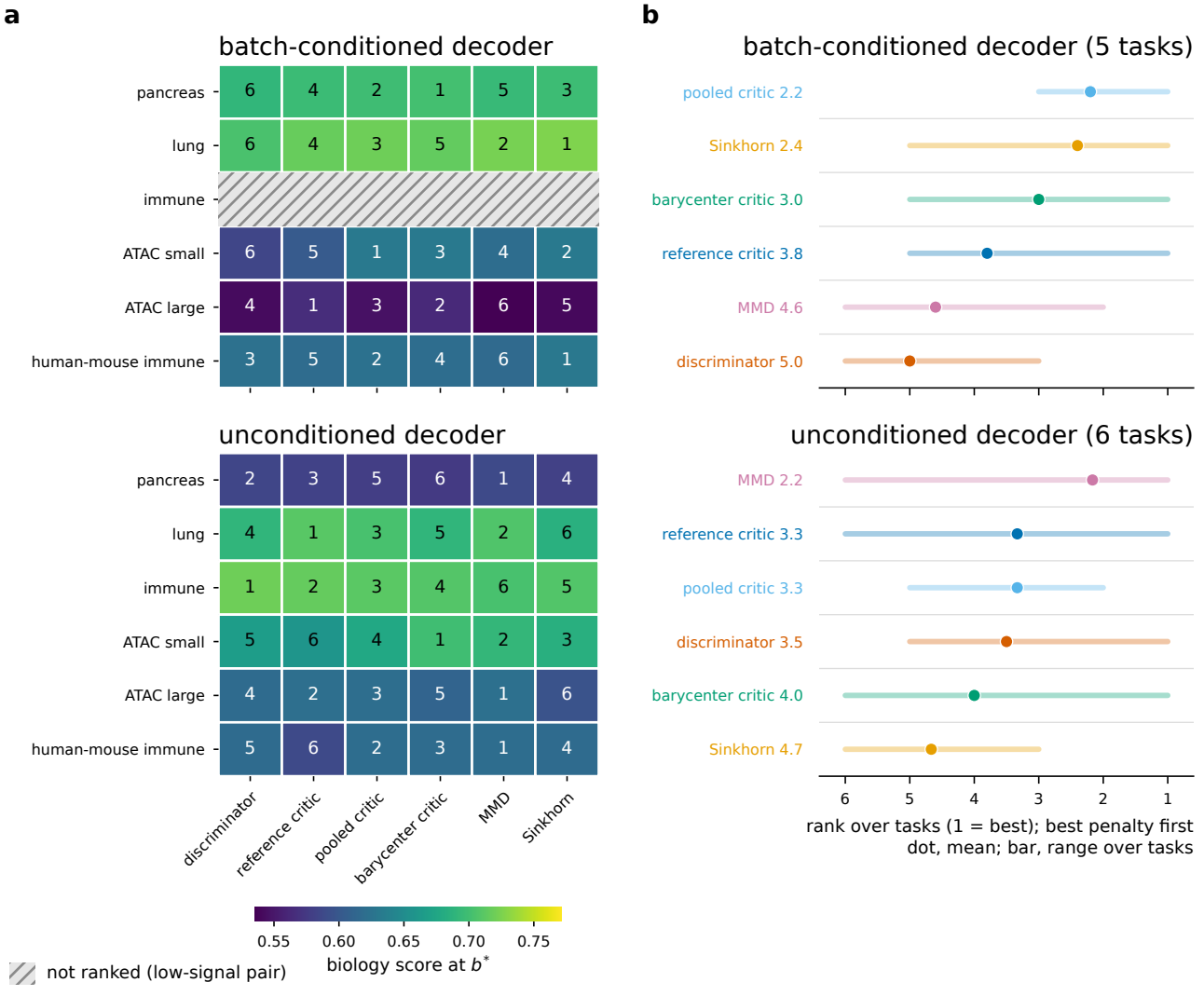

a
batch-conditioned decoder
pancreas 6 4 2 1 5 3
lung 6 4 3 5 2 1
immune
ATAC small 6 5 1 3 4 2
ATAC large 4 1 3 2 6 5
human-mouse immune 3 5 2 4 6 1
unconditioned decoder
pancreas 2 3 5 6 1 4
lung 4 1 3 5 2 6
immune 1 2 3 4 6 5
ATAC small 5 6 4 1 2 3
ATAC large 4 2 3 5 1 6
human-mouse immune 5 6 2 3 1 4
discriminator
reference critic
pooled critic
barycenter critic
MMD
Sinkhorn
0.55 0.60 0.65 0.70 0.75
biology score at b^*
not ranked (low-signal pair)
b
batch-conditioned decoder (5 tasks)
pooled critic 2.2
Sinkhorn 2.4
barycenter critic 3.0
reference critic 3.8
MMD 4.6
discriminator 5.0
unconditioned decoder (6 tasks)
MMD 2.2
reference critic 3.3
pooled critic 3.3
discriminator 3.5
barycenter critic 4.0
Sinkhorn 4.7
6 5 4 3 2 1
rank over tasks (1 = best); best penalty first
dot, mean; bar, range over tasks